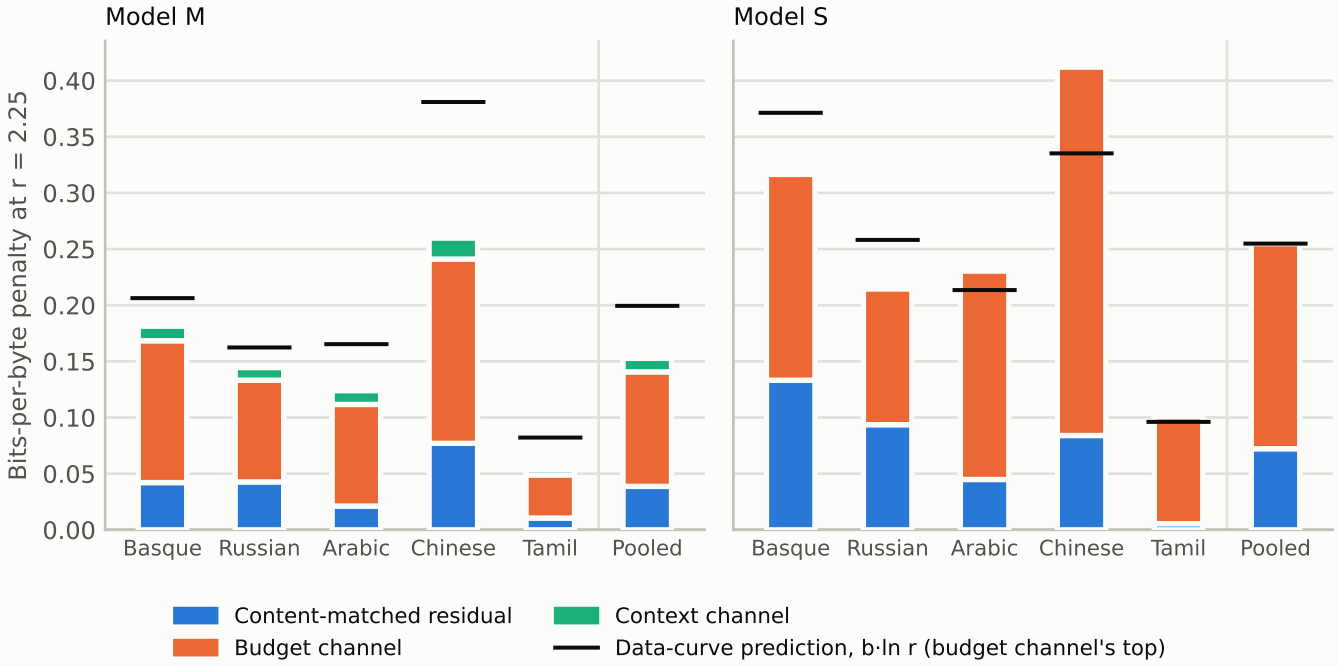Figure 2. Where the r = 2.25 penalty comes from